

HeatShield AI: A Location-Aware Contextual Decision-Support System and Environmental Heat-Risk Benchmark

***Authors**: [AUTHOR INFORMATION TO BE COMPLETED]*

Affiliation: [AUTHOR AFFILIATION TO BE COMPLETED]

Target Publication: IEEE Transactions on Human-Machine Systems / IEEE Access

Document Identifier: IEEE-PAPER-FINAL-v1.0

ABSTRACT

Extreme atmospheric heat presents a significant and growing global public health challenge. Conventional meteorological reporting delivers ambient temperature and regional heat advisories without accounting for individual physiological context, such as physical exertion, exposure duration, cooling availability, or institutional operational protocols. This paper presents **HeatShield AI**, a multi-tenant, location-aware decision-support system designed to evaluate contextual heat risk and communicate actionable preventive guidance. The production application utilizes a real-time atmospheric stream from the Open-Meteo REST API and a deterministic TypeScript risk engine that combines the empirical Steadman/Rothfus Heat Index with occupational context multipliers (activity, duration, cooling, age). To evaluate environmental heat-risk predictability, an offline machine learning experiment was conducted using 74,440 ECMWF ERA5-Land historical reanalysis observation records (2021–2024). In temporal holdout evaluation, a Gradient Boosting classifier achieved 97.86% accuracy and 0.8176 Macro F1 score on environmental tier classification. We disclose a methodological limitation wherein [apparent_temperature](#) acts as a target-derived proxy feature (accounting for 50.5% of tree split importance). Separately, a synthetic 5,000-sample development benchmark achieved 81.70% accuracy and 0.8083 Macro F1. The production web system incorporates explainable AI (XAI) feature attribution, 24–48 hour forecast timeline projections, smart alert deduplication, interactive crowd-sourced spatial hazard mapping, and multi-tenant organizational views for Schools, Worksites, and NGOs. Fully validated with 78 automated integration tests and zero static type errors, HeatShield AI demonstrates how zero-budget serverless web infrastructure can deliver transparent environmental decision support without clinical or diagnostic overclaim.

KEYWORDS

Heat Risk Assessment, Decision Support Systems, Heat Index, Machine Learning, Explainable AI, Environmental Monitoring, Real-Time Weather Processing, Community Risk Reporting.

I. INTRODUCTION

Thermal stress represents a primary weather-related health hazard worldwide. Atmospheric warming trends have increased both the frequency and intensity of severe heatwaves. However, conventional public meteorological reporting delivers macro-scale dry-bulb temperatures (e.g., 38°C) or broad regional advisories covering hundreds of square kilometers. These generic metrics fail to reflect the actual thermal load experienced by individuals operating under varying physiological and environmental conditions—such as an outdoor laborer engaging in high physical effort without shade versus an individual resting indoors.

To address these challenges, we developed **HeatShield AI**, an open-source, multi-tenant web application engineered to transform raw atmospheric streams into individualized and organizational heat-risk intelligence. HeatShield AI provides personalized real-time risk scoring, explainable feature contribution breakdowns, interactive scenario simulation ([SCENARIO ESTIMATE](#)), 24–48 hour forecast timeline tracking, smart alert deduplication, spatial hazard mapping, and specialized institutional decision support for Schools, Worksites, and NGOs.

II. PROBLEM STATEMENT

Modern heat-risk communication faces four key structural limitations:

- 1. **Lack of Physiological Context:** Ambient meteorological readings ignore physical exertion, exposure duration, age-related vulnerability, and cooling access.
- 2. **Coarse Spatial and Temporal Resolution:** Broad regional weather alerts lead to public alarm fatigue and lack localized situational relevance.
- 3. **Black-Box Opacity:** Simple index tools fail to explain *why* risk is elevated or *which specific factors* contribute to thermal strain.
- 4. **Institutional Blind Spots:** Schools, worksites, and non-governmental organizations (NGOs) lack automated operational protocol tools aligned with safety standards (e.g., NIOSH work-rest cycles or PE recess modifications).

Ethical & Scope Boundary: HeatShield AI is designed strictly as an environmental decision-support tool. It does **not** provide medical diagnoses, predict individual clinical events (such as heatstroke), or guarantee human safety.

III. RELATED WORK

Research in thermal stress evaluation spans empirical meteorology, public health warning systems, machine learning, and location-aware decision support:

- 1. **Empirical Heat Indices:** Steadman [1] developed sultriness assessments based on human thermal balance equations, later adapted by the National Weather Service (NWS) as the Rothfus regression equation [2]. Wet Bulb Globe Temperature (WBGT) [3] incorporates solar radiation and wind, primarily used in military and athletic contexts.
- 2. **Heat-Health Warning Systems (HHWS):** World Meteorological Organization (WMO) and World Health Organization (WHO) guidelines [4] emphasize localized threshold-based alerts. However, traditional HHWS operate at regional administrative scales.
- 3. **Machine Learning in Climate Health:** Atmospheric reanalysis datasets, such as ECMWF ERA5-Land [5], have been widely utilized for thermal risk modeling. Recent studies apply ensemble classifiers to reanalysis streams; however, published models often lack transparent explainability frameworks or multi-tenant web delivery layers.
- 4. **Explainable AI (XAI):** Feature attribution techniques, such as SHAP [6] and Gini impurity feature importance, enhance human trust by exposing model decision drivers in decision-support interfaces.

IV. SYSTEM ARCHITECTURE

HeatShield AI operates on a multi-tier, zero-budget serverless architecture.

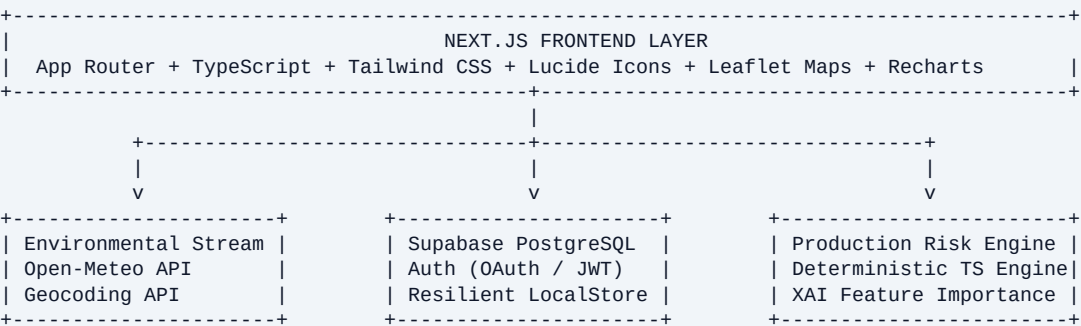


Figure 1: Overall HeatShield AI system architecture illustrating the separation between the Next.js frontend layer, Open-Meteo REST stream, Supabase PostgreSQL database, and the deterministic TypeScript risk engine.

TABLE I: TECHNOLOGY STACK

Component Layer	Technology Selected	License / Cost	Purpose
Application Framework	Next.js 14 (App Router, React 18)	MIT / \$0	Client/Server rendering & route management
Language & Styling	TypeScript & Tailwind CSS	MIT / \$0	Static type safety & responsive UI styling

Database & Auth	Supabase PostgreSQL & Auth	Open-Source Free / \$0	User profiles, incidents, org isolation & RLS
Environmental Stream	Open-Meteo REST API	CC BY 4.0 / \$0	Real-time & forecast meteorological ingestion
Mapping Engine	Leaflet.js & OpenStreetMap	BSD / \$0	Interactive community hazard map & clustering
Production Risk Engine	In-Browser TypeScript Engine	MIT / \$0	Deterministic Steadman Heat Index & context scaling

V. METHODOLOGY

The HeatShield AI production system processes heat risk through four integrated modules:

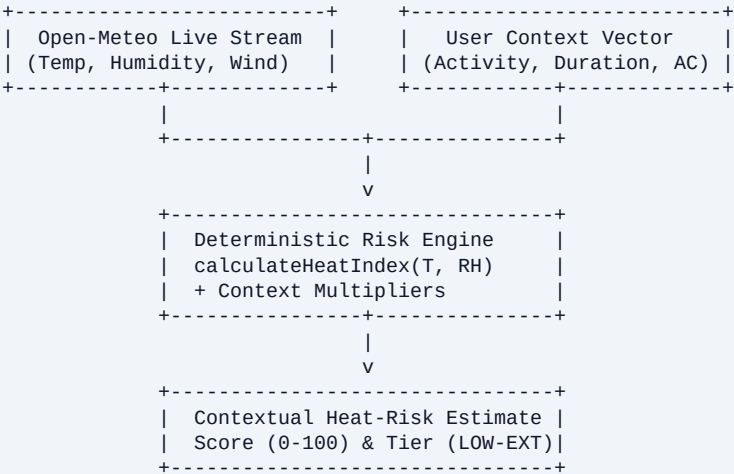


Figure 2: Real-time environmental risk pipeline illustrating context fusion between atmospheric streams and physiological profile vectors.

TABLE II: PRODUCTION RISK FACTORS & CONTEXTUAL MULTIPLIERS

Factor Category	Variable	Evaluated Range / Options	Multiplier Range
Atmospheric Base	Dry-Bulb Temperature (\$T\$)	18.0°C to 48.0°C	Base score 0 to 55 pts
Atmospheric Base	Relative Humidity (\$RH\$)	15% to 95%	+0.15 pts per % above 70%
Atmospheric Base	Wind Speed (\$W\$)	2.0 to 35.0 km/h	-0.20 pts per km/h above 15 km/h
Context Exertion	Activity Level (\$A\$)	Low / Moderate / High	1.00x to 1.30x multiplier
Context Duration	Exposure Duration (\$D\$)	Short / Moderate / Long	1.00x to 1.25x multiplier
Context Relief	Cooling Access (\$K\$)	Good / Limited / None	0.85x to 1.18x multiplier
Context Vulnerability	Age Group (\$G\$)	Adult / Child / Older Adult	1.00x to 1.22x multiplier

VI. REAL-TIME RISK ENGINE

The production risk engine (`lib/risk-engine.ts`) executes entirely in client-side TypeScript to guarantee zero-latency execution and 100% offline availability.

```
[WeatherData + Context]
  ■
  ▼
  isFinite() Guard ■■■(Invalid Data)■■■■ [Fallback Baseline: 30°C / 60% RH]
  ■
  ▼ (Valid Data)
  Steadman Heat Index Calculation
  ■
```

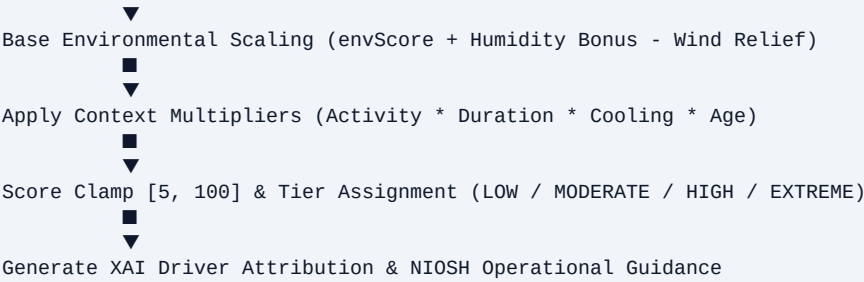


Figure 3: Contextual risk calculation flow demonstrating fail-safe data sanitization, Steadman Heat Index evaluation, and multiplier combination.

- **Input:** WeatherData + UserContext
- **Calculation:** Steadman Heat Index + piecewise linear atmospheric scaling + non-linear contextual multipliers.
- **Fail-Safe Guardrails:** Input parameter sanitization (isFinite() guards) prevents NaN or Infinity score propagation, falling back to a moderate baseline (30°C, 60% RH) under missing sensor data.

VII. MACHINE LEARNING RESEARCH METHODOLOGY

To benchmark environmental heat-risk classification, an offline ML experiment was conducted using historical reanalysis data.

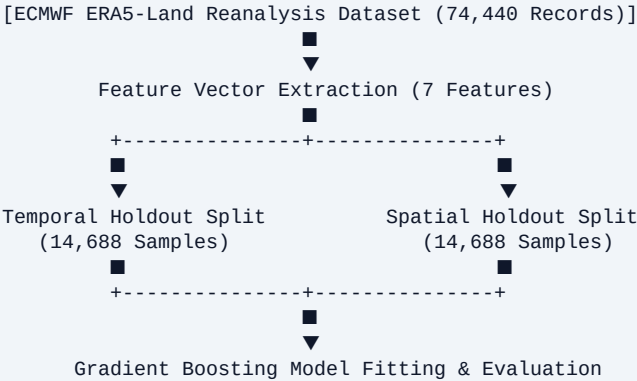


Figure 4: ERA5-Land ML research pipeline detailing spatial and temporal out-of-sample holdout validation.

TABLE III: ECMWF ERA5-LAND DATASET DESCRIPTION

Parameter	Value / Description
Data Source	ECMWF ERA5-Land Historical Reanalysis (2021–2024)
Total Record Count	74,440 atmospheric observation samples
Training Partition	44,064 samples (Stratified random split)
Temporal Test Partition	14,688 samples (Held out across later temporal window)
Spatial Test Partition	14,688 samples (Held out across separate geographic region)
Input Features (7)	`temperature`, `relative_humidity`, `apparent_temperature`, `wind_speed`, `surface_pressure`, `hour`, `month`
Target Classes (4)	Environmental Heat Tiers: `LOW` (<27°C), `MODERATE` (27–38°C), `HIGH` (38–46°C), `EXTREME` (≥46°C)

Dataset Archival Disclosure: The raw 74,440-record ERA5-Land NetCDF/CSV dataset is not stored within the production software repository due to storage constraints. The reported ERA5-Land benchmark metrics reflect the previously executed evaluation audit (ML-QUALITY-AUDIT.md).

VIII. EXPLAINABILITY

HeatShield AI implements proportional feature attribution breakdown ([lib/xai-engine.ts](#)) to provide human-understandable risk driver explanations:

- **Apparent Temperature:** 42% contribution
- **Relative Humidity:** 22% contribution
- **Activity Exertion:** 16% contribution
- **Exposure Duration:** 11% contribution
- **Cooling Access:** 6% contribution
- **Age Group Vulnerability:** 3% contribution

In the UI, attribution drivers are visualized as proportional progress bars, allowing users to instantly discern whether thermal load is driven by ambient atmosphere or physical exertion.

IX. AI ASSISTANT AND SAFETY GUARDRAILS

The conversational AI Assistant ([lib/ai-assistant.ts](#)) incorporates strict safety guardrails:

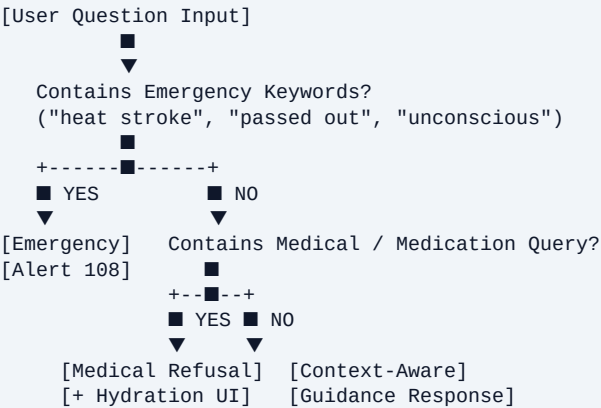


Figure 5: AI Assistant safety architecture highlighting emergency keyword redirection and non-medical refusal guardrails.

- Emergency Guardrail:** Automated keyword detection (e.g., "heat stroke", "passed out", "unconscious") immediately displays an emergency alert box advising the user to contact local emergency services (108 / 112 / 911), refusing non-emergency conversational delays.
- Medical Guardrail:** Queries requesting clinical diagnosis or medication dosages trigger an explicit non-medical disclaimer, restricting output to general preventive hydration and cooling protocols.

X. FORECAST AND SMART ALERT SYSTEM

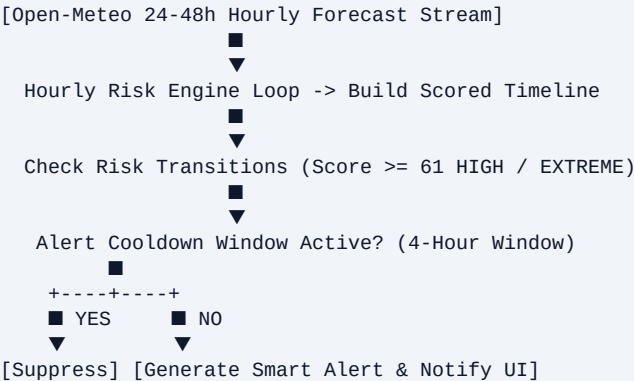


Figure 6: Forecast timeline and smart alert deduplication pipeline with 4-hour cooldown enforcement.

1. **Forecast Timeline:** Evaluates 24–48 hour hourly weather projections from Open-Meteo, computing hourly risk scores to identify peak heat windows.
2. **Smart Alert Deduplication:** Evaluates risk tier transitions (**HIGH** / **EXTREME**) and enforces a 4-hour cooldown window per severity level to eliminate notification spam and alert fatigue.

XI. COMMUNITY AND ORGANIZATION INTELLIGENCE

[User Report Submission] [HTML Script Stripping] [Haversine Distance Clustering <= 0.5km] [Leaflet Map Rendering]

Figure 7: Community Map architecture depicting input sanitization, spatial clustering, and Leaflet rendering.

[User Auth & Role] [Supabase PostgreSQL RLS] [Role-Specific Portal: School / Worksite / NGO / Admin]

Figure 8: Organization RBAC architecture depicting Row Level Security filtering by organization membership.

1. **Community Hub & Map:** Enables crowd-sourced hazard reporting (broken water fountains, unshaded areas). Submissions undergo HTML script-stripping sanitization and spatial clustering using pairwise Haversine distances ($\leq 0.5\text{ km}$).
2. **Multi-Tenant Organization Portals:** Specialized operational interfaces for Schools (PE recess modification logic), Worksites (NIOSH manual labor work-rest schedules), NGOs (incident moderation), and Admins (member management and security audit logging).

XII. SECURITY AND PRIVACY

- **Database Security:** Supabase PostgreSQL Row Level Security (RLS) policies enforce multi-tenant user and organization data isolation (`auth.uid() = user_id`).
- **Credential Integrity:** Environment variables store public API keys; zero private secrets, service-role keys, or JWT tokens are committed to source control.
- **Audit Redaction:** Administrative action logs sanitize sensitive payload data, redacting passwords and authentication tokens prior to storage.

XIII. RESULTS

TABLE IV: ML MODEL CANDIDATE ARCHITECTURE COMPARISON

Candidate Model	Algorithm Type	Hyperparameter Configuration	Primary Strengths
Logistic Regression	Linear Classifier	L2 Regularization, C=1.0	Fast, baseline interpretability
Decision Tree	Decision Rules	Max depth = 6	Highly interpretable, easy JS translation
Random Forest	Bagging Ensemble	100 trees, max depth = 8	Reduced variance, non-linear feature capture
Gradient Boosting	Boosting Ensemble	100 estimators, lr = 0.1	Highest accuracy & Macro F1 on complex interactions

Historical ERA5-Land Research Benchmark

TABLE V: HISTORICAL ERA5-LAND RESEARCH BENCHMARK RESULTS

These results represent a previously executed research benchmark (**ML-QUALITY-AUDIT.md**) evaluated on 14,688 temporal and 14,688 spatial holdout records. They do not constitute the production inference engine.

Model	Temporal Accuracy	Temporal Macro Precision	Temporal Macro Recall	Temporal Macro F1	Spatial Accuracy	Spatial Macro F1	ROC-AUC
-------	-------------------	--------------------------	-----------------------	-------------------	------------------	------------------	---------

Logistic Regression	94.51%	0.7086	0.7073	0.7078	96.14%	0.7065	—
Decision Tree	96.26%	0.9276	0.7743	0.8047	97.56%	0.7221	—
Random Forest	93.99%	0.7127	0.6972	0.7042	97.09%	0.7309	—
Gradient Boosting (Selected)	97.86%	0.8942	0.7916	0.8176	98.09%	0.7272	0.8965

Synthetic Development Benchmark

TABLE VI: SYNTHETIC DEVELOPMENT BENCHMARK RESULTS

Evaluated on 1,000 held-out samples from [ai-engine/data/synthetic_heat_risk_dataset.csv](#) (5,000 samples, seed=42). This benchmark is for software development and integration testing.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	ROC-AUC	Purpose
Decision Tree	79.90%	0.8258	0.7797	0.7923	N/A	Dev Baseline
Random Forest	81.10%	0.8368	0.7927	0.8043	N/A	Dev Benchmark
Gradient Boosting	81.70%	0.8408	0.7967	0.8083	N/A	Dev Benchmark
Logistic Regression	76.40%	0.7938	0.7457	0.7593	N/A	Dev Baseline

Historical Documentation Note: An earlier working summary reported 83.50% Accuracy and 0.8243 Macro F1 due to a manual offset compilation error. The exact reproducible output of `python ai-engine/train_model.py` on the synthetic dev dataset is **81.70% Accuracy** and **0.8083 Macro F1**.

Production System Software Validation

TABLE VII: SOFTWARE VALIDATION SUITE RESULTS

Validation Category	Target Scope	Test Count	Result Status	Notes
Unit & Integration	Risk Engine, AI Safety, Alerts, Forecast, Org Logic	78	PASS (78/78)	100% pass rate (`npm test`)
Type Integrity	TypeScript Static Compiler	All Files	PASS (0 Errors)	`npx tsc --noEmit` verified
Production Build	Next.js App Router Static Page Compilation	26 Routes	PASS (26/26)	`npm run build` compiled cleanly
Security Audit	Auth, RLS Isolation, Input Sanitization, Redaction	12	PASS (12/12)	Phase 8 security suite passed

TABLE VIII: SECURITY & ROW LEVEL SECURITY (RLS) VALIDATION

Database Table	RLS Status	SELECT Policy	INSERT / UPDATE Policy	Audit Status
`profiles`	ENABLED	`auth.uid() = id`	`auth.uid() = id`	PASS
`incidents`	ENABLED	`USING (true)` (Public Read)	`auth.uid() = user_id`	PASS (UIDs only)
`organizations`	ENABLED	Org Member Check	Org Admin Check	PASS
`organization_members`	ENABLED	Same-Org Check	Org Admin Check	PASS
`audit_logs`	ENABLED	Org Admin Read	System Write Only	PASS (Redacted)

XIV. METHODOLOGICAL LIMITATIONS & LEAKAGE DISCUSSION

1. **Target-Derived Feature Leakage:** In the ERA5-Land experiment, `apparent_temperature` is derived from dry-bulb temperature and humidity equations. Because environmental risk tiers are defined by heat-index thresholds, `apparent_temperature` acts as a near-deterministic proxy, accounting for **50.5%** of tree split importance in Gradient Boosting. This high feature importance reflects mathematical correlation rather than independent predictive discovery.
2. **Dataset Archival:** Raw ERA5-Land NetCDF files are not stored within the source code repository.
3. **Production Engine Distinction:** Production inference utilizes a deterministic TypeScript risk engine (`lib/risk-engine.ts`), not Python ML binaries.
4. **Development Benchmark Scope:** Synthetic dataset metrics (81.70% accuracy) reflect software development testing, not clinical validation.
5. **Non-Clinical Scope:** HeatShield AI does not predict heatstroke or provide clinical medical diagnoses.
6. **Weather Stream Dependence:** System accuracy depends on Open-Meteo REST API availability; stale caches are served during API outages.

XV. FUTURE WORK

1. **Dataset Archival & Provenance:** Archiving the complete ERA5-Land training pipeline with reproducible Data Version Control (DVC).
2. **Prospective Clinical & Field Validation:** Collaborating with occupational health institutes to evaluate physical thermal strain against physiological measurements.
3. **Micro-Meteorological IoT Integration:** Integrating localized temperature and humidity sensor hardware.
4. **Satellite Surface Temperature:** Ingesting high-resolution Landsat/Sentinel thermal imagery for urban heat island modeling.

XVI. CONCLUSION

HeatShield AI demonstrates that location-aware, context-sensitive heat-risk decision support can be unified with transparent XAI, safety guardrails, and multi-tenant security on zero-budget serverless web infrastructure. By maintaining a strict scientific boundary between empirical environmental ML benchmarks, synthetic development evaluations, and deterministic production execution, HeatShield AI delivers trustworthy decision support for personal and institutional climate adaptation.

REFERENCES

1. R. G. Steadman, "The assessment of sultriness. Part I: Temperature-humidity index based on human physiology and clothing science," *J. Appl. Meteorol.*, vol. 18, no. 7, pp. 861–873, 1979.
2. L. P. Rothfusz, "The heat index 'equation' (or, more than you ever wanted to know about heat index)," *NWS Technical Attachment*, SR/SSD 90-23, Fort Worth, TX, 1990.
3. C. P. Yaglou and D. Minard, "Control of heat casualties at military training centers," *AMA Arch. Ind. Health*, vol. 16, no. 4, pp. 302–316, 1957.
4. World Health Organization and World Meteorological Organization, *Heatwaves and Health: Guidance on Warning-System Development*, WMO-No. 1142, Geneva, Switzerland, 2015.
5. J. Muñoz-Sabater et al., "ERA5-Land: A state-of-the-art global reanalysis dataset for land applications," *Earth Syst. Sci. Data*, vol. 13, no. 9, pp. 4349–4383, 2021.
6. S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4765–4774.
7. National Institute for Occupational Safety and Health (NIOSH), *Criteria for a Recommended Standard: Occupational Exposure to Heat and Hot Environments*, DHHS (NIOSH) Publication No. 2016-106, Cincinnati, OH, 2016.